Database Management Systems: Foundations & Relational Model

Chapter 1: Relational Data Model and Key Constraints

A relational database organizes data into one or more tables (or relations) consisting of rows (tuples) and columns (attributes). Each table has a designated primary key that uniquely identifies each tuple within that table. A candidate key is a minimal superkey that can uniquely distinguish records. A primary key is chosen from candidate keys. Foreign keys establish referential integrity between tables, ensuring that values in the child table reference valid primary key tuples in the parent table.

Relational algebra forms the theoretical foundation of relational queries, comprising fundamental operations such as Selection (sigma), Projection (pi), Cartesian Product (cross), Union (cup), Difference (-), and Join (bowtie).

Chapter 2: Database Normalization and Normal Forms

Understanding Normalization (1NF, 2NF, 3NF, BCNF)

Normalization is a systematic approach of decomposing tables to eliminate data redundancy and undesirable anomalies such as Insertion, Update, and Deletion anomalies. It organizes relational database fields and tables according to functional dependencies.

First Normal Form (1NF) mandates that table columns contain only atomic (indivisible) values and each row contains unique values with no repeating groups. Second Normal Form (2NF) requires that the table is in 1NF and all non-key attributes are fully functionally dependent on the entire primary key, eliminating partial dependencies. Third Normal Form (3NF) requires 2NF compliance and ensures that no non-prime attribute is transitively dependent on the primary key (X -> Y and Y -> Z where Z is non-prime). The benefits of 3NF include minimal storage redundancy, simplified data consistency, and prevention of anomalous modifications. Boyce-Codd Normal Form (BCNF) is a stricter version of 3NF where for every functional dependency X -> Y, X must be a super key.

Chapter 3: Transaction Management & ACID Properties

ACID Guarantees and Concurrency Control

A transaction is a logical unit of database processing that includes one or more database access operations. To preserve database integrity, transactions must satisfy the four ACID properties: Atomicity (all or nothing execution), Consistency (database transitions from one valid state to another), Isolation (concurrent transactions execute independently without interference), and Durability (committed changes persist permanently even after hardware crashes).

Concurrency control techniques like Two-Phase Locking (2PL) and Timestamp Ordering ensure serializability and prevent dirty reads, unrepeatable reads, and phantom reads.